Mathematics: analysis and approaches
Standard level
Paper 1 — markscheme
Topic 4 Statistics — cumulative frequency

5 November 2026

W4ss Academy practice set 2 | **Topic** Topic 4 Statistics — cumulative frequency | **Ref.** W4SS–MAA–QB–P1–02

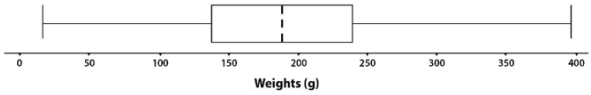
Student name

Paper 1

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- **M**: method marks. **A**: answer or accuracy marks, usually after the M mark they depend on. **R**: reasoning marks. **N** (former course): marks for a correct answer with no working.
- Marks in brackets, e.g. **(M1)**, can be implied by correct working that follows. **AG**: answer given in the question, no mark for it. **FT**: follow through from an earlier error.
- Total: **[19 marks]**.

Section A

Question	Marks	Markscheme
1. (a)	**[1]**	9 (cm) ***(A1)(C1)***
1. (b)	**[1]**	40 (leaves) ***(A1)(C1)***
2. (a)	**[1]**	130 employees ***A1 N1***
2. (b)	**[1]**	£320 ***A1 N1***
3. (a)	**[2]**	760 − 420 = 340 (g) ***(M1)A1***
3. (b)(i)	**[1]**	Median = 190 (g) ***A1***
3. (b)(ii)	**[1]**	Lower quartile = 135 − 140 (g) ***A1***
3. (b)(iii)	**[1]**	Upper quartile = 242 − 247 (g) ***A1***
3. (c)	**[2]**	 ***M1A1*** **Note:** The M1 is for a box and whisker plot and the A1 for all 5 statistics in the right places.

Section B

Question	Marks	Markscheme
4. (a)	**[1]**	75 ***A1***
4. (b)	**[2]**	recognition that all entries add up to 120 ***(M1)*** $a = 120 - 6 - 13 - 26 - b$ OR $a = 75 - b$ ***A1***
4. (c)(i)	**[3]**	$\frac{6\times1+13\times2+26\times3+(75-b)\times4+b\times5}{120} = 3.65$ ***(M1)(A1)*** **Note:** Award (M1) for attempt to substitute into mean formula, LHS expression is sufficient for the M mark. Award (A1) for correct substitutions in one variable OR in two variables, followed by evidence of solving simultaneously with $a + b = 75$. $b = 28$ ***A1***
4. (c)(ii)	**[2]**	120 − their part (c)(i) seen (e.g. 92 indicated on graph) ***(M1)*** 84 ***A1***